$$A_i \frac{\mathrm{d}c_i}{\mathrm{d}t} = -R_i A_i c_i - J_i, \qquad R_i = \frac{\rho_2 P_i}{A_i} + \frac{1}{T_{2B}}, \quad i \in \{S, B\}$$

$$\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2} - R_t u, \qquad R_t = \frac{\rho_2}{h} + \frac{1}{T_{2B}}, \qquad u(0) = c_S,\ u(L) = c_B$$

$$U(x) = \frac{c_S \sinh q(L-x) + c_B \sinh qx}{\sinh qL}, \qquad J_S = Dwq\left(c_S \coth qL - c_B \operatorname{csch} qL\right), \qquad q^2 = \frac{R_t - \lambda}{D}$$

$$\left[(R_S-\lambda)A_S S+DwC\right]\left[(R_B-\lambda)A_B S+DwC\right]-(Dw)^2=0,\quad S=\frac{\sinh qL}{q},\ C=\cosh qL$$

$$A_n = \frac{\left(\int_\Omega \phi_n \,\mathrm{d}\Omega\right)^2}{|\Omega| \int_\Omega \phi_n^2 \,\mathrm{d}\Omega}, \qquad \int_\Omega \phi_n = A_S c_S + A_B c_B + w \int_0^L U \,\mathrm{d}x$$

$$G_x = \frac{Dwq}{\sinh qL}, \qquad G_s = Dwq \tanh \frac{qL}{2}, \qquad q \simeq \sqrt{R_t/D} \quad (\lambda \ll R_t)$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} m_S \\ m_B \end{pmatrix} = -\begin{pmatrix} R'_S + k_S & -k_B \\ -k_S & R'_B + k_B \end{pmatrix}\begin{pmatrix} m_S \\ m_B \end{pmatrix}, \qquad R'_i = R_i + \frac{G_s}{A_i}, \quad k_i = \frac{G_x}{A_i}$$

$$qL \ll 1: \quad G_x \to \frac{Dw}{L}, \qquad G_s \to \tfrac{1}{2}\, wLR_t$$

$$qL \gg 1: \quad G_x \to 2Dwq\, e^{-qL} \to 0, \qquad G_s \to w\sqrt{DR_t}$$

$$\lambda_{f,s} = \frac{a + d \pm \sqrt{(a-d)^2 + 4k_S k_B}}{2}, \qquad a = R'_S + k_S, \quad d = R'_B + k_B$$

$$A_f = \frac{\bar{R}' - \lambda_s}{\lambda_f - \lambda_s}, \qquad \bar{R}' = p_S R'_S + p_B R'_B, \qquad A_s = 1 - A_f$$

$$\Psi = \frac{Dw/L}{A_S(R_S - R_B)}, \qquad \Psi_{\text{eff}} = \frac{k_S}{R'_S - R'_B}, \qquad \eta = \frac{A_f}{p_S}$$

$$\Psi_{\text{eff}} \to 0: \quad \lambda_f \to R'_S, \quad \lambda_s \to R'_B, \quad A_f \to p_S$$

$$\Psi_{\text{eff}} \gg 1: \quad \lambda_s \to \bar{R}', \quad \lambda_f \to \frac{k_S}{p_B}, \quad A_f \approx \frac{p_S p_B^3}{\Psi_{\text{eff}}^2}$$

$$R_{\text{eff}} = \frac{2\rho_2}{\bar{R} - 1/T_{2B}} = \frac{2(A_S + A_B)}{P_S + P_B} = \frac{r_B^2 + r_S^2}{r_B + r_S}$$

$$r = \frac{2\rho_2}{1/T_{2\mathrm{lm}} - 1/T_{2B}}$$

$$p_S = \frac{r_S^2}{r_S^2 + r_B^2} \approx f_S \implies r_S = r_B \sqrt{f_S / f_B}$$

$(r_S, r_B, G) \leftarrow (\lambda_f, \lambda_s, A_f)$ 两点交换（无汇）

$$\min_{r_S, r_B} \left\| (\ln \lambda_f, \ln \lambda_s, \ln A_f)_{\mathrm{TL}} - (\cdot)_{\mathrm{NNLS}} \right\|, \quad w, L \text{ 已知}$$

$A_S = 78.540,\ P_S = 30.414,\ A_B = 1256.64,\ P_B = 124.664 \Rightarrow R_S = 2.270 \times 10^{-3},\ R_B = 8.294 \times 10^{-4}\ \mathrm{ms}^{-1}$

$$R_t = \frac{0.005}{0.5} + \frac{1}{3000} = 0.01033, \qquad q = \sqrt{R_t/D} = 0.07188\ \mu\mathrm{m}^{-1}, \qquad qL = 1.0804$$

$$G_x = \frac{2.0 \times 1 \times 0.07188}{\sinh 1.0804} = 0.1103, \qquad G_s = Dwq \tanh \frac{qL}{2} = 0.07090 \ \mu\mathrm{m}^2/\mathrm{ms}$$

$$R'_S = 2.270 \times 10^{-3} + \frac{0.07090}{78.54} = 3.172 \times 10^{-3}, \quad R'_B = 8.858 \times 10^{-4}, \quad k_S = 1.404 \times 10^{-3}, \quad k_B = 8.778 \times 10^{-5}$$

$$\Psi_{\text{eff}} = \frac{k_S}{R'_S - R'_B} = \frac{1.404 \times 10^{-3}}{2.286 \times 10^{-3}} = 0.6143 \quad (\boxtimes\ \text{II})$$

$$\lambda_{f,s} = \frac{5.550 \times 10^{-3} \pm 3.671 \times 10^{-3}}{2} \Rightarrow T_{2,f} = 216.9 \text{ ms}, \quad T_{2,s} = 1064.2 \text{ ms}$$

$$\bar{R}' = 1.020 \times 10^{-3}, \qquad A_f = \frac{\bar{R}' - \lambda_s}{\lambda_f - \lambda_s} = 0.0220, \qquad \eta = 0.373$$

TL: $T_{2,f} = 228.8$ ms, $A_f = 0.0225$; NNLS: $T_{2\mathrm{lm}} = 214.8$ ms, 0.0231

$$\Psi_{\text{eff}} = 28.37, \qquad A_f \approx \frac{p_S p_B^3}{\Psi_{\text{eff}}^2} = 6.09 \times 10^{-5} \quad (\text{TL } 5.50 \times 10^{-5})$$

$$T_{2,f} \approx \frac{p_B A_S}{G_x} = \frac{0.9412 \times 78.54}{3.1728} = 23.3 \text{ ms} \quad (\text{TL } 23.0,\ \text{NNLS } 15.7 \text{ ms})$$

$$r_{\text{偽}} = \frac{0.01}{1/15.70 - 1/3000} = 0.16\ \mu\text{m} \quad (\text{真値}\ 5\ \mu\text{m})$$

$$r_S = \frac{0.01}{1/214.79 - 1/3000} = 2.31\ \mu\text{m}, \qquad r_B = \frac{0.01}{1/1086.72 - 1/3000} = 17.04\ \mu\text{m}$$

$$r_S = 18.76 \times \sqrt{\frac{0.0495}{0.9505}} = 4.28 \ \mu\text{m}$$

$$R_{\text{eff}} = \frac{0.01}{1/1121.08 - 1/3000} = 17.90\ \mu\text{m} \quad \text{vs.} \quad \frac{400 + 25}{25} = 17.00\ \mu\text{m}$$